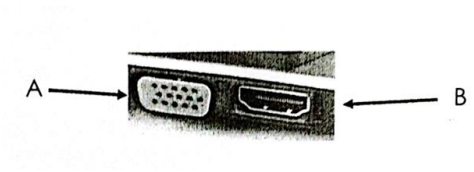
Higher National Diploma in Engineering (Electrical)

First Year, Second Semester Examination — 2021

EE 1212-Introduction to Computer Systems

Question 01 (Total 25 Marks)

I. Picture of two ports available in a laptop computer is given below.



a. Name the ports shown by A and B. (04 Marks)
b. Name two (02) devices that can be connected through the ports A and B. (02 Marks)

ii). Relevant to a CPU, what does it meant by $1.80\mathrm{GHz}$ (03 Marks)

iii). Briefly explain the purpose of having expansion slots in a motherboard. (04 Marks)

iv). What is **BIOS**? Briefly describe how BIOS works in a PC. (06 Marks)

v). Briefly explain the following computer memory types. (06 Marks) a) Registers b) Cache memory

Question 02 (Total 25 Marks)

i). Briefly explain the difference between **Address bus** and **Control bus**. (04 Marks)

ii). Describe the following software categories. (06 marks) a. Open source software b. Commercial software

iii). Write down the features of **Linux** operating system. (04 Marks)

iv). List two (02) examples for each type of software given below. (08 Marks) a. Translator software b. General purpose Application software c. Special purpose Application Software d. Virus guards

v). Write down the meaning of following units relevant to a printer. (03 Marks) a. dpi b. ppm c. cps

Question 03 (Total 25 Marks)

i). Briefly explain the functionality of a **web browser**. (04 Marks)

ii). What is a **firewall**? Mention the advantage of adding a firewall to a network. (06 Marks)

iii). Expand the abbreviation **ISP** and briefly explain the duty of ISP. (04 Marks)

iv). Propose suitable application software for performing the following tasks. (05 Marks) a. Preparing monthly budget b. Recolor a faded photograph c. Writing a resignation letter d. Preparing a presentation e. Storing students' personal details and exam results

v). Briefly describe the network devices given below. (06 Marks) a. Repeater b. Switch c. Router

Question 04 (Total 25 Marks)

i). Briefly explain the following terms regarding networks. (06 marks) a. Client-server model b. Peer to peer model

ii). What is **spyware**? Explain how to prevent yourself from spyware. (05 Marks)

iii). What is a **Blog**? (02 Marks)

iv). Convert the following binary numbers into decimal equivalent. (06 Marks) a) 1000.10 b) 110111.001 c) 11.1100

v). Perform the following operation using **two's complement** notation. (06 Marks) a) $-25 + 38$ b) $-29 - 35$

Question 05 (Total 25 Marks)

i). What is a **Leased line connection**? Write down two advantages and two disadvantages of Leased line connection. (04 Marks)

ii). Give two example for **network protocol**. (02 Marks)

iii). Briefly describe the following methods for connecting to the Internet. (04 Marks) a. ADSL b. SDSL

iv). A company owner needs to have a network within the office building and provide Internet connection to all the employees. Currently company has the following resources.

- Server computer
- 13 Desktop computers
- One Network Printer
- Existing ADSL connection
- One Fax machine

The network needs to satisfy following requirements.

- All desktop computers are allowed to access Internet and that will be control by server.
- Printer and Fax machine should be accessed by all desktop computers.
- Internet connection will be given to the employees through their smart phones.

a) What is the type of the cable most suitable for connecting desktop computers? (02 Marks) b) Mention the most suitable type of network topology for this network. Give reason to your answer. (04 Marks) c) Name a device to distribute internet connection to smart phones. (02 Marks) d) Propose an operating system for the server computer. (02 Marks) e) Draw a network diagram for the above network and name the components. (05 marks)